Relaytic-AML: A Local-First Evaluation Lab for Financial-Crime ML

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June 2026

Abstract

Financial-crime machine learning is usually judged by detector scores, but operational AML work depends on evidence a team can trust: data posture, temporal validity, graph provenance, review capacity, and claim discipline. Relaytic-AML is a local-first evaluation lab for that setting. It keeps the workspace as the source of truth, records data and modeling decisions as inspectable artifacts, and lets specialist agents help without making their prose the authority.

Relaytic began as a general local inference-engineering system. In this work it is sharpened into an AML-focused environment where a human or external agent can ask what is known, what is blocked, and what should happen next. The public evidence includes supporting PaySim synthetic temporal-fraud PR-AUC 0.638773, supporting Elliptic temporal graph-feature PR-AUC 0.668756, and Elliptic2 modern-context PR-AUC 0.94324 +/- 0.000882, below the recorded RevClassifyDS reference of 0.974. The contribution is not a detector-superiority claim. It is a reproducible architecture for keeping experiments, review assumptions, limitations, and reported claims aligned.

1 Introduction

AML systems are operational decision systems. A useful model does not only rank transactions or entities. It has to respect time, explain why a case reached a review queue, preserve graph or entity provenance, and make clear what a result does not prove. This is especially hard in financial-crime work because the most realistic data is private, licensed, regulated, or operationally sensitive.

That privacy boundary creates a scientific problem as much as an engineering problem. Public AML research often has to work with synthetic data, anonymized transaction graphs, or partial blockchain views, while real production teams care about delayed labels, changing behavior, investigation budgets, typologies, and model-risk review. A benchmark row is therefore incomplete unless the reader can see the data contract, the split rule, the operating point, and the reason a stronger interpretation is or is not allowed.

Relaytic was first built as a broader local inference lab for structured data. The AML edition is a deliberate narrowing of that idea. Instead of trying to be a generic benchmark runner, Relaytic-AML asks a more practical question: can a local machine or local server become a trustworthy evaluation lab where data, modeling work, agent assistance, review assumptions, and public claims stay connected?

The answer explored here is architectural. Relaytic-AML treats the local workspace as the authority. Models, tables, context packs, figures, and paper text are downstream of artifacts on disk. Agents are useful, but they act through bounded roles. A guide explains the current state. A scout checks data posture and leakage risk. A scientist challenges the modeling plan. A builder runs the experiment. A claim governor decides whether a result can be described publicly. The user is never expected to know which file to open first because Relaytic itself can explain the run state and export a compact context pack for another LLM or coding agent.

The benchmarks in this paper are deliberately modest in their role. They are stress tests for the environment. A score is valuable only if the reader can see the dataset boundary, split rule, budget, operating point, limitation, and claim boundary that produced it. This paper therefore presents Relaytic-AML as an evaluation lab first and a benchmark package second.

2 Contribution and Scope

This paper is a systems paper about local-first AML evaluation. Its first contribution is the Relaytic-AML architecture: a workspace-centered lab where artifacts, not chat transcripts, carry the truth of an experiment. The second contribution is a role model for agentic help that keeps guidance, scouting, scientific challenge, model building, and claim review separate. The third contribution is an evidence-cell contract that makes every reported number traceable to a dataset, split, command, artifact field, budget, operating point, and claim state. The fourth contribution is a claim discipline that keeps useful evidence from becoming stronger than it deserves to be.

This scope matters. Relaytic-AML does not claim hard AML superiority, production readiness, graph-neural superiority, or equivalence to RevClassify. It shows a working local evaluation environment that can run meaningful public evidence, expose where stronger claims remain blocked, and give both humans and agents a stable way to continue the work.

Paper claim	Relaytic mechanism	Current evidence posture
Local state is auditable	Run directories, manifests, traces, metric cells, tables, and release reports live in the workspace	Demonstrated by the generated paper and source package
Agent help is bounded	Guide, scout, scientist, builder, and claim governor roles have separate jobs and artifacts	Architecture evidence; not yet a formal autonomy benchmark
Scores are traceable	Evidence cells bind every number to dataset, split, command, artifact field, budget, and claim state	Supporting PaySim, Elliptic, and Elliptic2-context rows
Release claims are evidence-bounded	Metric cells, limitation records, and release audits prevent unsupported promotion	Hard and headline AML claims remain blocked

3 Problem Setting and Design Thesis

The central problem is legibility under constraint. A trained reader can understand PR-AUC, temporal splits, graph features, and review budgets, but that reader cannot know the state of a local experiment unless the system makes it legible. Relaytic-AML is built around the view that an evaluation environment should expose its state in the same way a model exposes its metrics: through stable artifacts, not through informal memory.

The design thesis is that local-first evidence can make agent-assisted AML research more reliable if three invariants hold. First, the local artifact graph must remain the authority for data posture, split decisions, metric values, model artifacts, and release state. Second, agent assistance must be role-scoped: a guide may explain, a scout may inspect, a scientist may challenge, a builder may execute, and a release governor may block a claim, but none of them should silently overwrite the evidence record. Third, every public interpretation of a result must be bounded by what the evidence cell can actually support.

This paper evaluates whether the current Relaytic-AML implementation makes those invariants concrete. The benchmark rows are useful because they exercise the architecture, not because they settle AML detection. The more important question is whether a human, a local model, or an external coding agent can enter the workspace, understand what is known, see what remains blocked, and choose a safe next action without private rows leaving the local boundary.

4 Local-First Agent Architecture

Relaytic is designed around one control rule: the user's workspace is the authority. Raw and licensed data stay local by default. Run directories, manifests, traces, metric cells, model files, and paper assets form the durable record. Semantic caches, memory indexes, and LLM summaries are derived views. This is different from a remote-first agent that sends private rows to a hosted planner and later reconstructs provenance from a conversation.

The system is agentic, but the roles are concrete. They are small jobs with bounded permissions and visible outputs. A scout inspects source posture and split risk. A scientist challenges baselines and ablations. A builder executes a controlled run. An evidence reviewer can reject an interpretation even when the model score looks attractive. A guide explains the current state to a human or exports a redacted context pack to another model.

Role	What it owns	Local-first boundary	Main artifacts
Operator and mandate owner	Sets goals, constraints, privacy posture, and stop/continue preferences.	Can keep all data on a controlled local machine, server, or cluster.	Mandate, policy, permission, and next-action artifacts.
Guide and assist layer	Answers where the run is, what artifacts matter, and which action is safe next.	Uses local artifacts first. Optional LLM help is advisory and redacted by default.	Guide payloads, assist turns, status, and context packs.
Scout and task-contract agents	Inspect source posture, target semantics, split validity, and leakage risk.	Work from staged local snapshots rather than mutating the original data source.	Dataset registry, source manifests, split contracts, and task reports.
Scientist and challenger agents	Propose baselines, ablations, shadow candidates, and failure explanations.	Candidate work is bounded by explicit budgets and local artifact permissions.	Experiment registry, scorecards, ablations, and shadow-trial reports.
Builder and search controller	Execute reproducible model/search plans and select thresholds on validation evidence.	Optional adapters are versioned and never become hidden sources of truth.	Run directories, model artifacts, search traces, and operating-point records.
Evidence and release governors	Decide how evidence may be described and which claims stay blocked.	Fail closed on leakage, missing provenance, unsupported interpretation, or dirty release state.	Metric cells, claim boundaries, release notes, and source-bundle audits.
External agents or LLMs	Consume exported context, propose repairs, or continue work through stable surfaces.	Receive rowless/redacted context unless policy explicitly grants richer access.	External context packs, handoff reports, and reproducible commands.

Two design choices carry most of the system. First, important work produces a local artifact that another human or agent can inspect. Second, optional intelligence is subordinate to the artifact graph. A local LLM may help phrase guidance, and a frontier model may suggest repairs, but neither becomes the source of truth unless its proposal is converted into a reproducible local artifact.

5 Current Frontier Context

The current AML frontier is not a single leaderboard. It is a set of pressure points: real graph scale, realistic entity behavior, temporal drift, operational throughput, and reliable agent-assisted research. Relaytic-AML is designed as infrastructure around those pressure points, not as a replacement for detector papers.

Recent agentic-ML work also treats external state, benchmark environments, and research validity as first-class objects. SkillOpt frames agent skills as trainable external state with validation-gated edits Yang et al. [2026]. MLR-Bench evaluates whether agents can produce valid open-ended ML research rather than only fluent reports Chen et al. [2025]. PaperBench

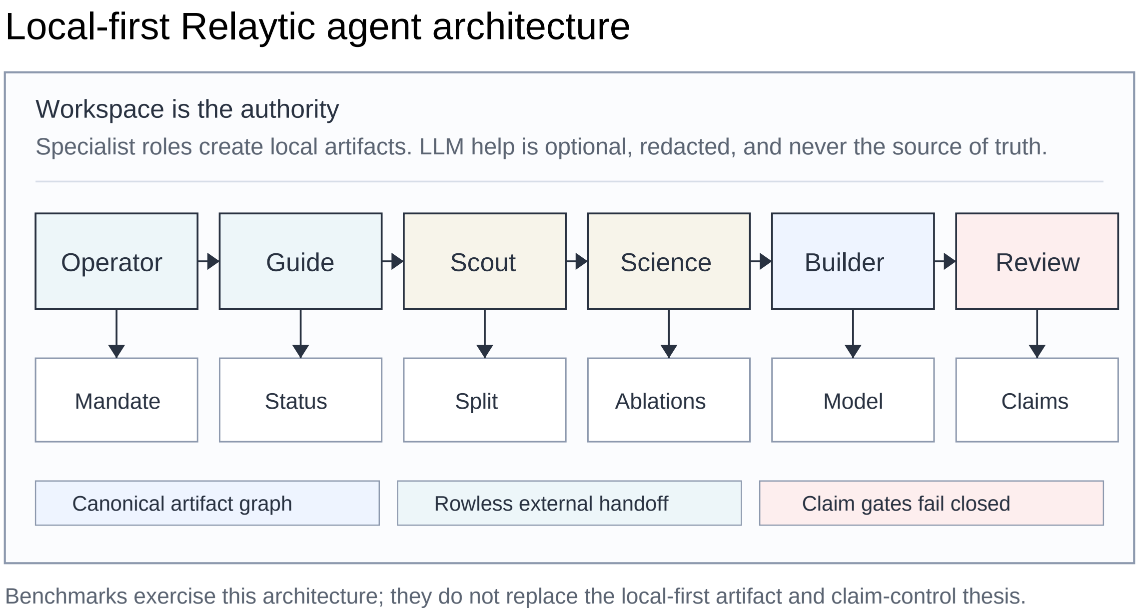

Figure 1: Local-first Relaytic agent architecture

studies whether agents can reproduce full ML papers from scratch and reports that the best tested agent remained far below expert human replication performance Starace et al. [2025]. Relaytic-AML follows the same broader movement toward executable research state, but applies it to AML-specific data custody, review queues, and release claims.

PaySim is a synthetic mobile-money simulator designed to address the scarcity of legitimate public mobile-transaction datasets for fraud research Lopez-Rojas et al. [2016]. It is useful for temporal transaction-fraud workflow evaluation, but synthetic evidence cannot by itself establish hard real-bank AML performance.

The Elliptic Bitcoin dataset introduced a public transaction graph with more than 200K transaction nodes, 234K directed payment-flow edges, and 166 node features across 49 time steps Weber et al. [2019]. That work also showed why graph evidence must be compared against strong simpler baselines rather than assumed superior.

Elliptic2 shifts the public AML benchmark center toward subgraph learning, with 121,810 labeled subgraphs inside a background graph of roughly 49M node clusters and 196M edge transactions Bellei et al. [2024]. RevTrack and RevClassify further argue that sender and receiver context around a subgraph can be a powerful and scalable signal Song et al. [2024]. These works motivate Relaytic-AML's modern-context and limitation track, but they do not make the current Relaytic Elliptic2 row a performance contribution.

Recent AML graph work raises the bar beyond the current Relaytic evidence rows. TransXion frames benchmark realism around profile-aware simulation, richer entity attributes, non-template illicit synthesis, and out-of-character behavior Chen et al. [2026]. LineMVGNN and ExSTraQt focus on directed money flow, edge-aware graph views, and quasi-temporal transaction representations Poon et al. [2026] Tariq and Hassani [2026]. BlazingAML treats throughput and multi-stage graph mining as a systems problem Ye et al. [2026]. Continual graph-learning reviews emphasize drift, adaptation, class imbalance, and changing laundering behavior Deprez et al. [2025]. These papers point to a frontier where realism, scale, graph structure, time, and operations are inseparable. Relaytic-AML is complementary to those efforts: it does not claim detector parity with them, but tries to make dataset posture, split validity, budget, limitations, and release claims auditable.

The paper also follows broader ML documentation and reproducibility practice. Datasheets for Datasets and Model Cards argue for explicit dataset and model reporting Gebru et al. [2021] Mitchell et al. [2019]. The NeurIPS reproducibility program highlights the need for code, data, and checklist discipline in ML research Pineau et al. [2021]. Recent work on ML research agents warns that coherent papers can still contain invalidated experiments, which reinforces the need for executable artifacts, reproducible commands, and explicit claim boundaries Chen et al. [2025] Starace et al. [2025].

6 Methodology: Evidence Cells, Gates, and Budgets

Relaytic-AML treats a metric as an evidence cell, not as a free-standing score. An evidence cell records the dataset identity, split contract, execution command, run or artifact reference, metric value, budget tier, leakage posture, operating-point rule, claim state, and limitation notes. A reported row is accepted only when those fields are present.

This creates a small theory of evidence for the system. A metric becomes scientific evidence only after it is attached to provenance, a comparison budget, and an interpretation boundary. Provenance answers where the number came from. Budget answers how much modeling effort was spent before reporting it. The interpretation boundary answers what the number is allowed to mean in public. Removing any one of those pieces turns the row back into an anecdote.

The claim gate is intentionally conservative. It can mark a row as supporting evidence, modern context, baseline-only evidence, or blocked evidence. In the current public version, no row is headline-eligible. This is not a weakness of the environment. It is the point of the environment. A strong-looking number is only useful if the system can also say what the number is allowed to mean.

The budget ladder separates quick engineering checks from serious evidence. Smoke runs test that commands and artifacts exist. Baseline runs establish conservative rows. Competitive runs add stronger feature families, validation-only threshold selection, calibration, and model search. Frozen reporting runs preserve the transformation from experiment to table, figure, and public claim. This prevents a weak first pass from being quietly promoted into a strong paper claim.

For an external reader, the important idea is simple: every number has a trail, and every claim has a boundary. The trail helps another person reproduce or challenge the result. The boundary says whether the result is a benchmark observation, an operational estimate, a limitation, or a claim that should not be made yet.

7 Evidence Operating Layer

The local-first architecture becomes concrete through an evidence operating layer. Relaytic-AML is not only a model runner. It coordinates source posture, task semantics, split discipline, model search, decision thresholds, review queues, artifacts, and claim boundaries.

Layer	What Relaytic records	Why it matters
Source and task contracts	Dataset access, target semantics, benchmark posture, and split rules	A reader can see whether the task is valid before judging the score
Execution and search	Candidate families, budgets, calibration, thresholds, and selected runs	Model development becomes inspectable instead of anecdotal
AML domain layer	Entity graphs, typology posture, review queues, delayed labels, and case evidence	The model is tied to the analyst workflow it is supposed to support
Evidence ledger	Metric cells, tables, figures, limitations, and commands	Numbers are connected to the artifacts that produced them
Claim boundaries	Allowed claims, blocked claims, and future unlock conditions	The same result cannot quietly become a stronger interpretation
Handoff surfaces	Guide, status, assist, mission control, and redacted context export	Humans and external agents can continue work without guessing hidden state

This operating layer is useful even when a detector claim is blocked. A blocked row is not thrown away. It becomes a structured research state with a reason, an artifact reference, and a repair path.

8 What Relaytic-AML Is For

The intended user is a person or team that has data, a risky modeling question, and a need to know whether the evidence is strong enough to act on. That includes a bank team comparing a new model against an incumbent queue, a fraud group exploring a new dataset, a researcher testing a benchmark protocol, or an external agent trying to continue a run without seeing private rows.

Capability	Reader-facing meaning
Local-first privacy posture	Private or licensed data can stay outside the public repo while hashes, access posture, and evidence artifacts remain inspectable
Artifact-first reproducibility	Tables and figures are tied to local JSON and model artifacts rather than loose notes
Role-scoped agent help	A guide, scout, scientist, builder, and claim reviewer have different responsibilities
Budget-aware modeling	Quick checks, baseline evidence, competitive runs, and frozen reporting are kept separate
Operational evaluation	Review-budget precision, recall, false-positive burden, and case-packet completeness sit next to model metrics
External context export	Another LLM or coding agent can receive a redacted summary and reproducible commands

This is why the project moved from a broad Relaytic system toward Relaytic-AML as the flagship story. The general architecture still matters, but AML gives it a sharper test. It forces the system to handle rare events, temporal splits, graph context, human review limits, privacy, and public-claim discipline at the same time.

Companies could use this kind of lab to challenge incumbent rules or models on the same review queue, evaluate whether a new dataset is worth deeper investment, audit whether a vendor comparison is fair, and prepare evidence packs for compliance or model-risk review. Researchers could use it to ask whether an AML result is supported, blocked, or mainly useful as a limitation.

9 Evaluation Environment

The current evaluation environment combines public benchmark evidence with local artifact discipline. Dataset registry artifacts define source and access posture. Split contracts define chronological, graph-snapshot, or subgraph partition rules. Benchmark runners produce model and operating-point artifacts. Tables and figures then read from those artifacts.

The current public evidence uses three tracks. PaySim is a synthetic temporal proxy. Elliptic is temporal graph-feature evidence. Elliptic2 is retained as modern subgraph context and limitation evidence because a faithful RevClassify reference-protocol reproduction has not yet been completed locally.

The environment follows three design rules. Local artifacts are the source of truth. Validation selects models, thresholds, and operating points before fixed test evaluation. Blocked evidence stays visible because hiding failed or incomplete tracks makes both human and agent-assisted research less scientific.

10 Benchmark Protocol

The benchmark protocol is deliberately subordinate to the architecture. Its purpose is to check whether Relaytic can produce traceable evidence, respect split and leakage contracts, expose operational assumptions, and block unsupported claims. A single score does not define the value of the system.

The public rows separate baseline and competitive evidence. Competitive rows use stronger feature families, candidate search, calibration, and validation-only operating-point selection. The goal is not to pretend the current numbers are final. The goal is to make their scope visible.

The evidence summary below should be read as a claim-boundary table as much as a performance table. The numbers matter, but the posture column is what stops them from being overstated.

Evidence row	Metric	Value	Claim posture
PaySim baseline	test PR-AUC	0.331345	baseline-only
PaySim competitive	test PR-AUC	0.638773	supporting-only synthetic temporal proxy
PaySim competitive	precision at review budget	0.703336	supporting-only
PaySim competitive	recall at review budget	0.471584	supporting-only
Elliptic graph-feature	test PR-AUC	0.668756	supporting-only graph-feature evidence
Elliptic graph-feature	precision at review budget	1	supporting-only
Elliptic graph-feature	recall at review budget	0.056604	supporting-only
Elliptic2 context	official-partition PR-AUC mean	0.94324	modern context only
Elliptic2 context	official-partition PR-AUC std	0.000882	modern context only
RevClassifyDS reference	published PR-AUC	0.974	reference context, not parity

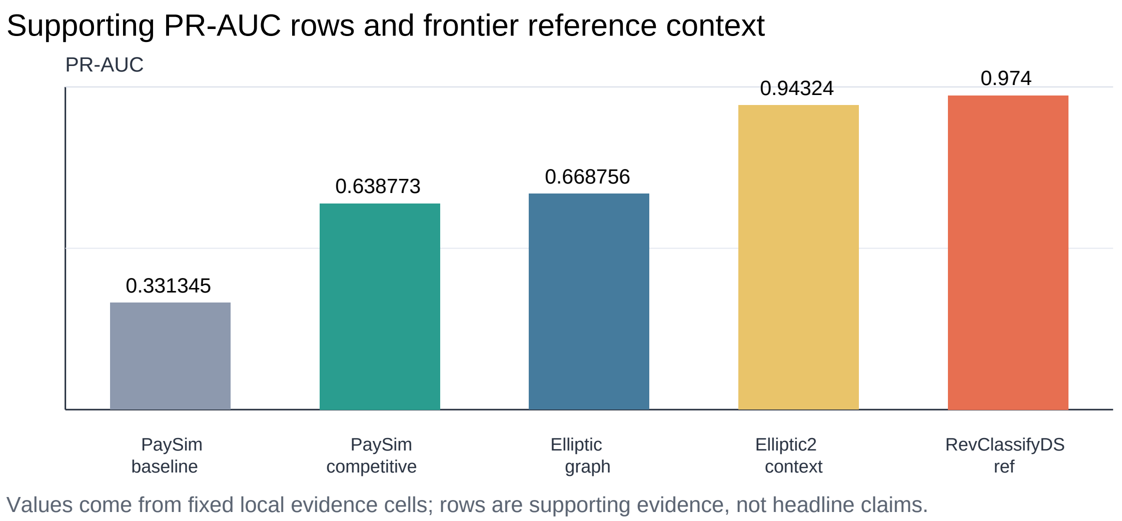

Figure 2: Supporting PR-AUC rows with claim posture

11 Results

The PaySim competitive row improves over the PaySim baseline inside the synthetic temporal-fraud contract. The Elliptic graph-feature row is useful supporting graph evidence, but it is not a graph-neural superiority claim. The Elliptic2 context row is strong enough to justify more work, but not enough to claim parity with the RevClassifyDS reference or to make an Elliptic2 performance contribution.

The more important result is the behavior of the environment. Relaytic can carry a useful score and still refuse the stronger sentence. In financial-crime ML, that refusal is not cosmetic. It is part of scientific and operational honesty.

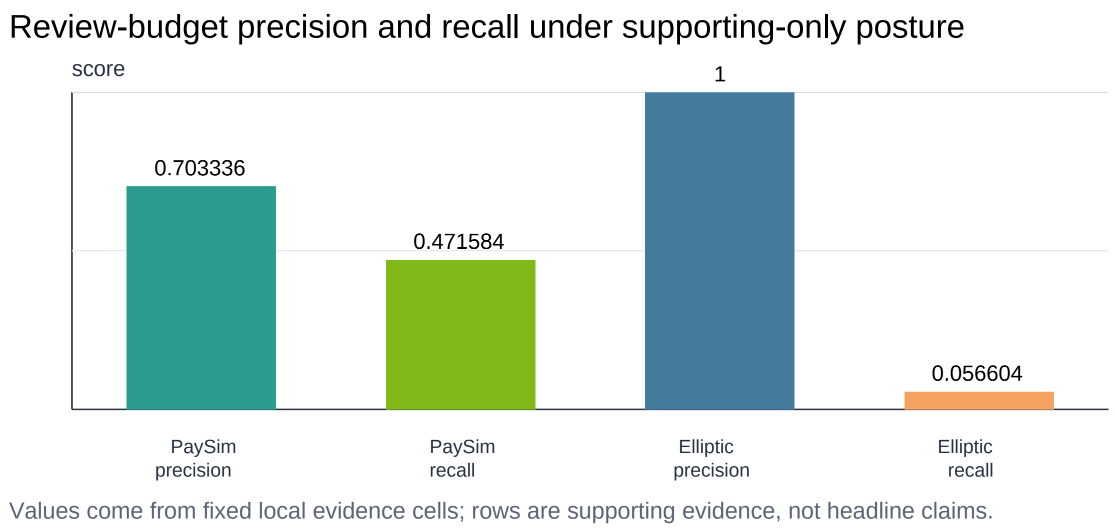

Figure 3: Review-budget precision and recall

Claim boundaries and future unlocks

Every track remains useful, but stronger claims require specific evidence upgrades.

Track	Current role	Blocked claim	Future unlock	Current posture
PaySim	synthetic temporal proxy	real-bank superiority	real/partner holdout	supporting
Elliptic	graph-feature support	graph-neural superiority	repeated graph release budget	supporting
Elliptic2	modern subgraph context	RevClassify parity	faithful parity + cohort proof	context only
Operational	review-budget support	hard business value	same-queue incumbent	supporting

Figure 4: Claim boundaries and future unlocks

Track	Current paper use	Blocked stronger claim	Evidence needed before promotion	Gate status
PaySim temporal proxy	supporting synthetic temporal-fraud evidence	real-bank AML superiority	partner or real holdout with frozen evaluation budget	supporting only
Elliptic graph-feature	supporting temporal graph-feature evidence	graph-neural or graph benchmark superiority	repeated graph evaluation budget against strong feature baselines	supporting only
Elliptic2 subgraph	modern subgraph context and limitation evidence	Elliptic2 performance contribution or reference-method match	faithful RevClassify reproduction or leakage-resistant subgraph protocol	context only
Operational review layer	supporting review-budget estimates	hard analyst-hour or business-value claim	complete case packets and same-queue incumbent comparison	supporting only

12 Discussion

The practical value of Relaytic-AML is not that it replaces a compliance platform or wins one benchmark table. Its value is that it gives risk, fraud, or AML teams a local evidence lab for unknown datasets, incumbent challenges, review-budget tradeoffs, leakage checks, and claim discipline. A team evaluating a new dataset can inspect whether the result is a model win, an environment win, a proxy-only result, or a blocked claim.

For research teams and agentic ML workflows, the same structure is a guard against coherent but invalid experiments. External agents can read the structured artifacts, see blocked claims, and propose the next benchmark action without inferring hidden state from prose. The strongest story is the artifact discipline: Relaytic-AML demonstrates that an ambitious evaluation system can still refuse claims it has not earned.

The system is also valuable when results are not spectacular. A low or blocked row can still tell a team that a dataset is synthetic-only, that a split leaks future information, that a graph-native candidate failed to beat strong tabular features, that a review-queue claim lacks a same-queue incumbent, or that a paper sentence is stronger than the evidence permits. In high-risk financial-crime settings, those negative answers are productive outputs.

An organization would use Relaytic-AML not as a replacement for domain expertise, but as an evaluation layer that makes difficult assumptions explicit: what data was allowed to move, what split was used, what budget was spent, whether the threshold was chosen on validation evidence, what a reviewer would see, and which interpretation the evidence still refuses to support.

13 Limitations

The current public evidence is not a deployment validation. PaySim is synthetic mobile-money evidence, so it cannot establish real-bank AML superiority. The Elliptic row is a temporal graph-feature result, not proof that a graph-neural model is better. Elliptic2 is modern context, not a Relaytic performance contribution, because faithful reference-protocol reproduction and cohort equivalence still need more work.

The operational evidence also remains early. The review-budget rows are useful for showing how Relaytic connects model output to analyst capacity, but they do not prove analyst-hour savings against an incumbent queue. A stronger version of this work should include complete case packets, a same-queue incumbent comparison, and a partner-approved or otherwise realistic holdout.

The paper therefore does not make a hard AML superiority claim. It argues that Relaytic-AML is a useful local evaluation environment and that the current benchmark rows are enough to demonstrate the architecture. They are not enough to end the detector question.

14 Future Work

The next step is to make the architecture more robust against misinterpretation and the evidence more operationally realistic. Relaytic itself should be evaluated as an object of study: whether the guide, scout, scientist, builder, and evidence reviewer agree about the same local run state; whether a first-time user can recover the right artifacts without knowing the repository; and whether an external model can use a redacted context pack to propose a useful repair without inventing unsupported claims.

The AML evidence should move in parallel. A stronger version needs a partner-approved or otherwise realistic holdout, faithful Elliptic2 reference reproduction, stronger graph-native candidates, continual-learning experiments, and same-queue business-value comparisons. The point is not to replace the current architectural story with a leaderboard story. The point is to give the architecture harder evidence to govern.

A useful next paper would therefore have two coupled evaluations. The detector evaluation would test stronger AML candidates under frozen budgets and leakage-resistant splits. The environment evaluation would test whether humans and agents can recover state, identify missing evidence, choose the right next action, and avoid unsupported claims. That would turn Relaytic's usability promise into a measurable benchmark rather than a narrative assertion.

The longer-term goal is still broader than AML. Relaytic should become a general local evaluation laboratory for structured, temporal, and graph ML. AML is the current flagship because it forces the system to handle privacy, time, graph context, human review, and claim discipline together.

15 Reproducibility

The code, paper source, figures, tables, and public evidence artifacts are in the Relaytic repository. The public repo keeps raw private or licensed data out of version control. Where a benchmark requires local data, the command ledger describes the expected local paths and access posture.

A compact reproduction path for the public paper assets is:

```
relaytic release-safety paper-tables --format json
relaytic release-safety paper-draft --format json
relaytic release-safety paper-release --format json
relaytic release-safety paper-arxiv-source --format json
relaytic scan-git-safety
```

The main reader-facing files are `docs/paper/relaytic_aml_arxiv_draft.md`, `docs/paper/relaytic_aml_arxiv_draft.pdf`, `docs/paper/arxiv_src/`, `docs/paper/figures/`, and `docs/paper/tables/`. The benchmark command ledger is `docs/reports/paper_reproduction_commands.md`. The README explains the current project shape: Relaytic remains the general package and CLI, while Relaytic-AML is the flagship AML edition used for this paper.

16 Author Use of AI Assistance

LLM-based tools assisted with drafting, editing, repository inspection, and consistency checks. The scientific framing, claim boundaries, experimental interpretation, and final manuscript remain the author's work. The tools are not listed as authors.

17 Conclusion

Relaytic-AML should be read as a local-first AML evaluation-lab paper. The current work is valuable because it makes the operating idea concrete. Data stays locally governed. Specialist roles create inspectable artifacts. External agents receive structured context instead of hidden state. Claim boundaries prevent stronger wording than the evidence has earned. The same evidence does not support hard AML superiority, a headline benchmark claim, graph-neural superiority, claimed equivalence to RevClassify, or hard business value. That restraint is not a footnote. It is part of the contribution.

The right test for this version is therefore not whether it ends the AML detector race. It does not. The right test is whether the repository makes the current evidence easier to inspect, easier to challenge, and harder to oversell. That is the claim this paper is prepared to make.

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